DATA WAREHOUSING & OLAP ANALYSIS

G.C.E. Advanced Level 2020 Dataset

Student Index No.	GAHDSE252F-006
Module	Data Warehousing
Database	PostgreSQL 18
Warehouse	AL_DataWarehouse
OLAP Tool	Google Colab / Python / Pandas
Fact Records	1,012,659

Day 04 – OLAP Cube Implementation

1. Introduction

This report presents the implementation of an Online Analytical Processing (OLAP) cube using the Sri Lankan G.C.E. Advanced Level 2020 examination dataset. The data warehouse was implemented using PostgreSQL and a star schema, while Google Colab, Python and Pandas were used to perform multidimensional OLAP analysis.

2. Data Warehouse Overview

The warehouse consists of a central fact table and several dimension tables. The grain of the fact table is one student's result for one subject in the examination.

Table	Records
fact_student_performance	1,012,659
dim_student	337,553
dim_subject	62
dim_stream	8
dim_syllabus	2
dim_exam	1
dim_gender	3

3. OLAP Cube Design

The OLAP cube was designed using three main dimensions: Stream, Subject and Syllabus. Average Z-score is used as the main analytical measure and result count is used as an additional measure.

Component	Description
Dimension 1	Stream
Dimension 2	Subject
Dimension 3	Syllabus
Measure 1	Average Z-score
Measure 2	Result Count

Cube structure: Stream × Subject × Syllabus → Average Z-score

4. OLAP Roll-up

Roll-up summarizes detailed data into a higher-level aggregation. In this implementation, subject-level information is aggregated to the stream level.

Result: Average Z-score by Stream and Syllabus.

5. OLAP Drill-down

Drill-down moves from a higher-level summary to a more detailed level. The analysis moves from Stream to individual Subject and calculates result count and average Z-score.

6. OLAP Slice

The Slice operation selects one value from a dimension. New Syllabus was selected, allowing Stream and Subject performance to be analyzed while keeping Syllabus fixed.

7. OLAP Dice

The Dice operation selects multiple values from multiple dimensions. The analysis selected the ARTS and COMMERCE streams, ACCOUNTING, ECONOMICS and BUSINESS STUDIES subjects, and New Syllabus.

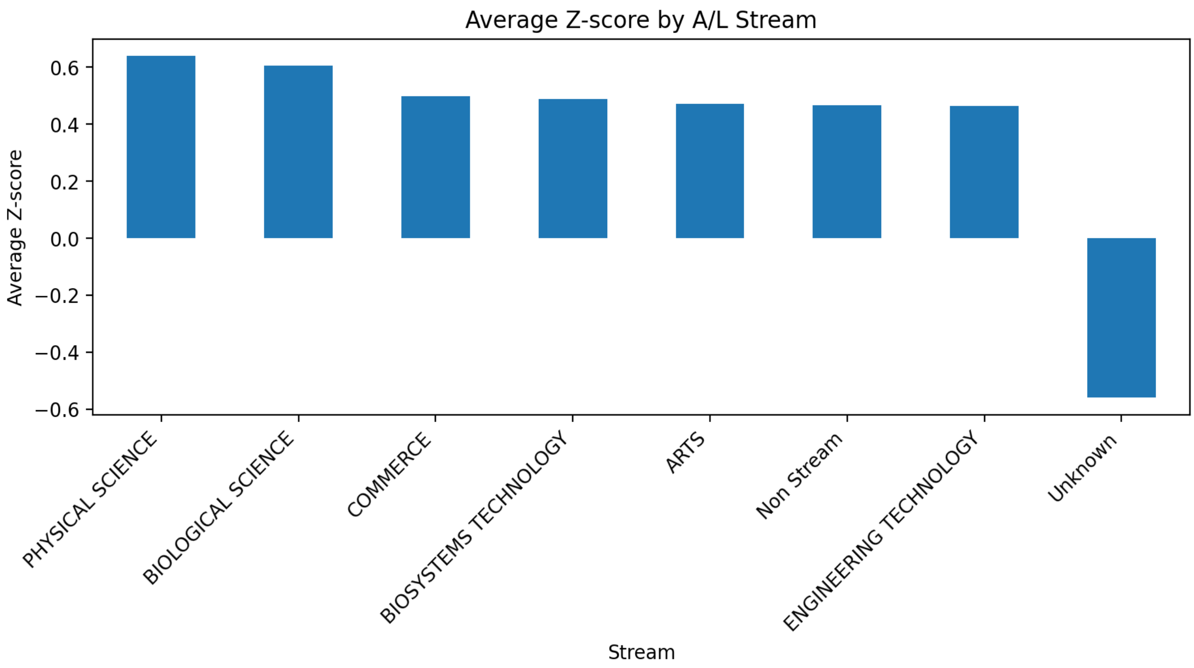
8. OLAP Pivot

The Pivot operation changes the orientation of the analytical view. Subject is placed on rows while Stream is placed on columns. The underlying data remains unchanged.

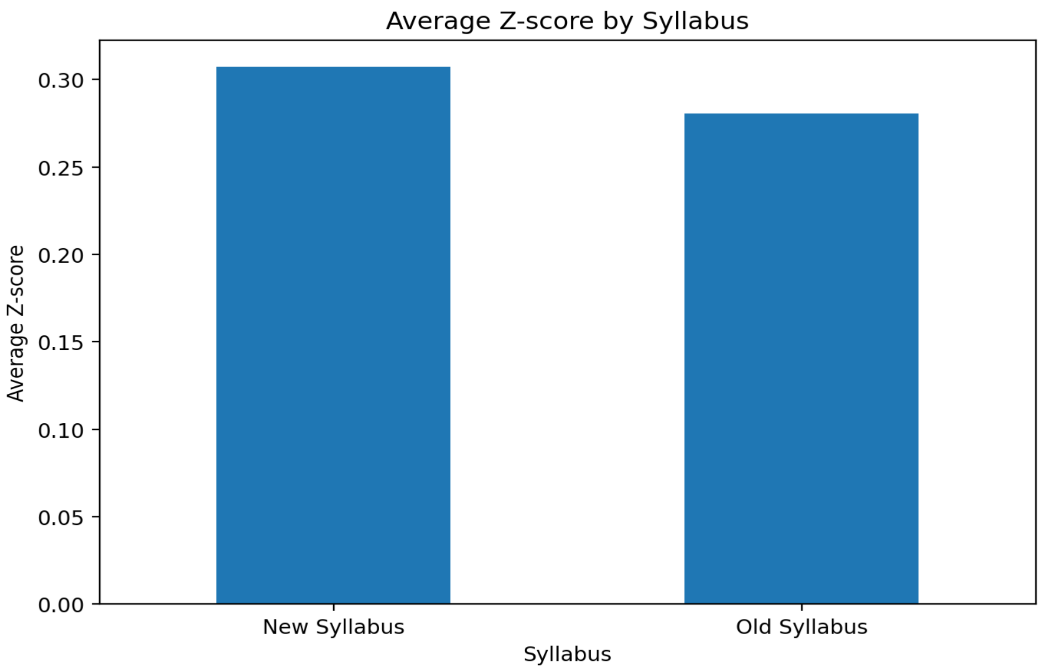
9. OLAP Visualizations

The following visualizations summarize important patterns identified through the OLAP analysis.

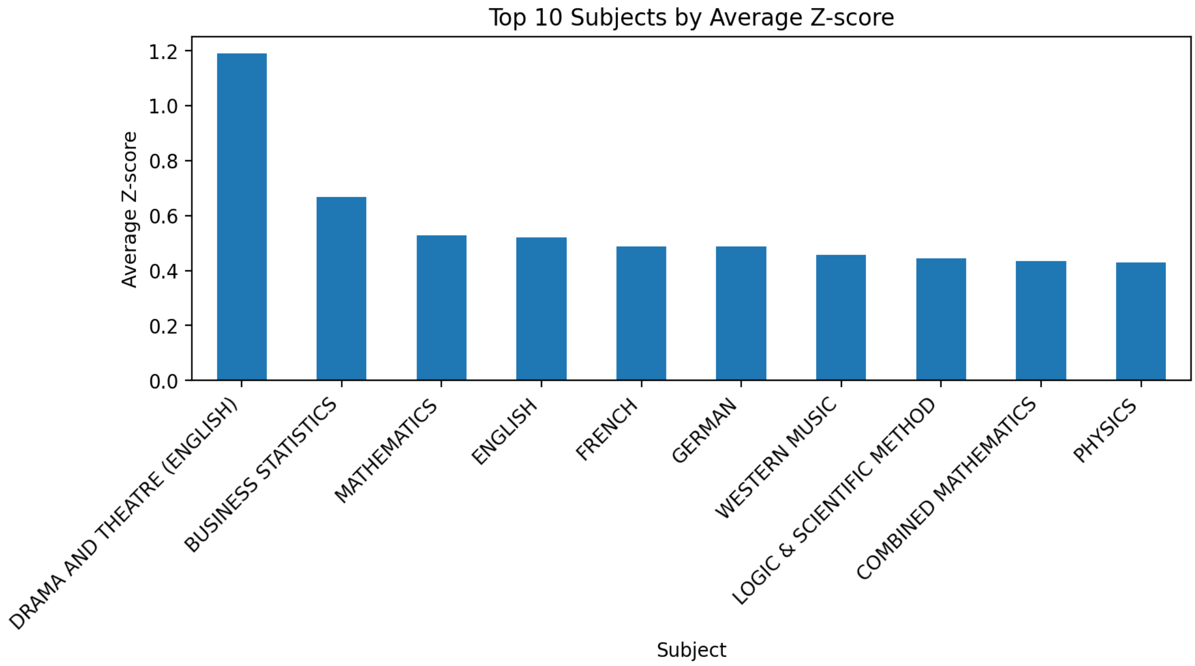
9.1 Average Z-score by Stream



9.2 Average Z-score by Syllabus



9.3 Top 10 Subjects by Average Z-score



10. Key Findings

• The stream with the highest average Z-score was **PHYSICAL SCIENCE**, with an average Z-score of **0.6380**.

• The highest average Z-score among the top 10 subjects was **DRAMA AND THEATRE (ENGLISH)**, with a value of **1.1911**.

• The OLAP cube contains 8 streams, 62 subjects and 2 syllabus types.

• Roll-up provides higher-level summaries, while drill-down provides more detailed subject-level analysis.

• Slice and Dice allow specific portions of the multidimensional data to be selected for focused analysis.

• Pivot allows the same analytical information to be viewed from a different dimensional orientation.

11. Conclusion

The OLAP implementation successfully transformed the PostgreSQL data warehouse into a multidimensional analytical environment. Stream, Subject and Syllabus were used as the main dimensions, while Average Z-score and Result Count provided analytical measures. The Roll-up, Drill-down, Slice, Dice and Pivot operations demonstrated how the examination dataset can be analyzed from different perspectives. The visualizations further supported the identification of performance patterns across streams, subjects and syllabus types.